Job Acceptance Prediction System

Introduction

The Job Acceptance Prediction System is a data science project focused on analyzing candidate profiles and predicting placement status using machine learning.

The project uses candidate academic scores, technical and aptitude scores, communication skills, work experience, and job-related attributes to understand the factors associated with placement outcomes.

The project combines Python, data analysis, machine learning, MySQL, and Streamlit to build an end-to-end analytical system.

Problem Statement

Organizations and placement teams need to understand which candidate characteristics are associated with successful placement outcomes.

Analyzing candidate data manually can make it difficult to identify important patterns and relationships.

This project aims to analyze historical candidate data and develop a machine learning classification model that predicts whether a candidate is classified as Placed or Not Placed.

Project Objectives

- Clean and prepare candidate data for analysis.
- Understand the distribution of placement outcomes.
- Analyze the relationship between candidate skills and placement status.
- Study the influence of academic performance and work experience.
- Examine job-role matching, company tier, and other job-related factors.
- Build and evaluate a machine learning classification model.
- Present key findings through an interactive Streamlit dashboard.

Dataset Description

The dataset contains 51,500 records and 26 columns before data cleaning.

The target column is status, which contains two categories: Placed and Not Placed.

Important dataset features

Feature	Description
ssc_percentage	Secondary school percentage
hsc_percentage	Higher secondary school percentage
degree_percentage	Degree percentage
technical_score	Candidate's technical assessment score
aptitude_score	Candidate's aptitude assessment score
communication_score	Candidate's communication assessment score
years_of_experience	Candidate's work experience
internship_experience	Internship-related information
expected_ctc_lpa	Expected compensation in lakhs per annum
relocation_willingness	Candidate's willingness to relocate
company_tier	Category or tier of the company
job_role_match	Match between candidate skills and job role
competition_level	Competition-related information
status	Target variable: Placed or Not Placed

Data Cleaning and Preprocessing

Data cleaning is performed to improve data quality before analysis and model training.

The main activities include:

1. Check the dataset structure and data types.
2. Identifying and handling duplicate records.
3. Identifying missing values.
4. Handling missing values using suitable methods based on the feature type and business meaning.
5. Standardizing inconsistent categorical values, such as differences in capitalization and extra spaces.
6. Check numeric columns for invalid values and unusual observations.
7. Preparing categorical and numerical features for machine learning.
8. Separating input features from the target variable.

The original dataset contains 1,376 duplicate rows and missing values in several columns. The final record count and preprocessing decisions should be documented after the cleaning process is finalized.

Exploratory Data Analysis (EDA)

EDA is used to understand the dataset, identify patterns, and explore relationships between candidate characteristics and placement outcomes.

Candidate Performance Analysis

- Compare academic scores between placed and non-placed candidates.
- Analyze technical, aptitude, and communication scores by placement status.
- Investigate whether stronger assessment scores are associated with placement.

Skills and Interview Performance

- Examine the relationship between job-role matching and placement status.
- Compare candidate assessment scores across placement groups.
- Identify patterns that may help explain placement outcomes.

Experience and Placement

- Analyze placement outcomes across experience levels.
- Compare candidates with and without internship experience.
- Examine whether work experience is associated with placement status.

Company Tier and Job-Related Factors

- Compare placement outcomes across company tiers.
- Examine expected compensation and placement status.
- Investigate the relationship between competition level and placement outcomes.

Correlation Analysis

A correlation heatmap can help identify relationships between numerical features.

Correlation indicates association, not necessarily causation. The analysis should not assume that one feature directly causes a candidate to be placed.

Machine Learning

Machine learning is used to learn patterns from historical candidate records and predict the target variable.

Problem Type

This is a **supervised binary classification problem** because the dataset contains a known target variable with two classes: Placed and Not Placed.

Input Features

The model can use suitable candidate attributes, such as:

- Academic scores
- Technical, aptitude, and communication scores
- Work and internship experience
- Job-role matching
- Company tier
- Other relevant job-related attributes

The final input features depend on the preprocessing and feature-selection steps.

Target Variable

The target variable is the status.

- Placed: The candidate is classified as placed in the dataset.
- Not Placed: The candidate is classified as not placed in the dataset.

The prediction target must follow the dataset's definition. Placement status should not be treated as a separate measurement of offer acceptance unless the source data explicitly establishes that meaning.

Model Training

The dataset is divided into training and testing sets.

- **Training set:** Used to teach the model patterns from historical data.
- **Testing set:** Used to evaluate how well the trained model performs on unseen records.

Data preprocessing should be fitted on the training data and then applied to the test data to avoid data leakage.

One or more classification algorithms can be evaluated, and the final model should be selected based on suitable evaluation metrics.

Model Evaluation

The model should be evaluated using:

- **Accuracy:** Percentage of predictions that are correct.
- **Precision:** How many candidates predicted as placed are actually placed.
- **Recall:** How many actually placed candidates are correctly identified.
- **F1-score:** Balance between precision and recall.

- **Confusion matrix:** Shows correct and incorrect predictions for both classes.

Accuracy alone may not be sufficient if the target classes are imbalanced.

The selected algorithm and actual evaluation scores should be added after model training is complete.

Streamlit Dashboard

The Streamlit dashboard provides an interactive interface for exploring candidate data and placement insights.

Suggested dashboard sections

Overview

- Total candidates
- Number of placed candidates
- Number of non-placed candidates
- Placement percentage

Candidate Analysis

- Academic scores versus placement status
- Technical, aptitude, and communication scores
- Skills match versus placement outcomes

Placement Analysis

- Placement rate by company tier
- Experience versus placement status
- Internship experience and placement
- Other relevant categorical comparisons

Machine Learning

- Model evaluation metrics
- Confusion matrix
- Feature importance, where supported by the selected model
- Placement-status prediction for new candidate inputs, if implemented

The dashboard should display values calculated from the actual dataset and trained model.

Tools and Technologies

Technology	Purpose
Python	Main programming language
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib and Seaborn	Data visualization
Plotly	Interactive charts
Scikit-learn	Model training and evaluation
Streamlit	Interactive dashboard
Jupyter Notebook	Experimentation and documentation

Expected Results

The project aims to:

- Identify important patterns in candidate placement data.
- Understand how candidate characteristics differ across placement outcomes.
- Build a classification model to predict placement status.
- Evaluate the model using appropriate classification metrics.
- Provide accessible visual insights through a dashboard.
- Demonstrate an end-to-end data science workflow.

Actual findings and model performance should be reported based on the final analysis and evaluation results.

Limitations

- Predictions depend on the quality and representativeness of the historical dataset.
- Missing or inconsistent values may affect analysis and model performance.
- Relationships found in historical data do not necessarily establish causation.
- A model's predictions are estimates and should not be treated as guaranteed outcomes.
- The model should be evaluated on unseen data before being used for practical decisions.

Conclusion

The Job Acceptance Prediction System demonstrates how data cleaning, exploratory data analysis, machine learning, and dashboard development can be combined to study placement outcomes.

By analyzing candidate profiles and evaluating a classification model, the project aims to provide useful insights into the factors associated with placement status and demonstrate the practical application of data science in recruitment and placement analytics.